

For Loop Diagnostic test: SOLUTIONS

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how for loops work is critical so if you have any problems we want to sort them out now.

You must show all your working in the following. No correct table (exactly the same)... no marks.

1. What is printed when the following code fragment has executed?

```
int a = 3; // L1
for (int i = 5; i <= 7; i++) // L2
{
    a = a + 7; // L3
    System.out.println(a); // L4
}
System.out.println(a+1); // L5
```

Printed:

10
17
24
25

LINE	a	i	i<=7 ?	Printed
L1	3			
L2		5	5<=7? true	
L3	10			
L4				10
L2		6	6<=7? true	
L3	17			
L4				17
L2		7	7<=7? true	
L3	24			
L4				24
L2		8	8<=7? false	
L5				25

2. What is printed when the following code fragment has executed?

```
int a = 3; // L1
for (int i = 5; i < 7; i++) // L2
{
    a = a + 7; // L3
    System.out.print(a); // L4
}
System.out.print(a+1); // L5
```

Printed:
101718

LINE	a	i	i<7 ?	Printed
L1	3			
L2		5	5<7? true	
L3	10			
L4				10
L2		6	6<7? true	
L3	17			
L4				17
L2		7	7<7? false	
L5				18

3. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int i = 0; i <= 1; i++) // L2
{
    System.out.println(a); // L3
    a = a + 5; // L4
}
System.out.println(a); // L5
```

Printed:

4
9
14

LINE	a	i	i<=1 ?	Printed
L1	4			
L2		0	0<=1? true	
L3				4
L4	9			
L2		1	1<=1? true	
L3				9
L4	14			
L2		2	2<=1? false	
L5				14

4. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int i = 0; i <= 2; i++) // L2
{
    a = a + 5; // L3
}
System.out.println(a); // L4
```

Printed:

19

LINE	a	i	i<=2 ?	Printed
L1	4			
L2		0	0<=2? true	
L3	9			
L2		1	1<=2? true	
L3	14			
L2		2	2<=2? true	
L3	19			
L2		3	3<=2? false	
L4				19

5. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int j = 0; j < 0; j++) // L2
{
    a = a + 5; // L3
}
System.out.println(a); // L4
```

Printed:
4

LINE	a	j	j<0 ?	Printed
L1	4			
L2		0	0<0? false	
L4				4

6. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int j = 0; j <=0; j++) // L2
{
    a = a + 5; // L3
    System.out.println(a); // L4
}
```

Printed:
9

LINE	a	j	j<=0 ?	Printed
L1	4			
L2		0	0<=0? true	
L3	9			
L4				9
L2		1	1<=0? false	

7. What is printed when the following code fragment has executed?

```
int b = 3; // L1
for (int i = 7; i >= 5; i--) // L2
{
    b = b + 7; // L3
    System.out.println(b); // L4
}
```

Printed:

10
17
24

LINE	b	i	i>=5 ?	Printed
L1	3			
L2		7	7>=5? true	
L3	10			
L4				10
L2		6	6>=5? true	
L3	17			
L4				17
L2		5	5>=5? true	
L3	24			
L4				24
L2		4	4>=5? false	

8. What is printed when the following code fragment has executed?

```
int b = 10; // L1
for (int i = 1; i < 5; i=i+2) // L2
{
    b = b + 10; // L3
    System.out.println(b); // L4
}
```

Printed:

20
30

LINE	b	i	i<5 ?	Printed
L1	10			
L2		1	1<5? true	
L3	20			
L4				20
L2		3	3<5? true	
L3	30			
L4				30
L2		5	5<5? false	